

Based on the latest CBSE blueprint, the syllabus for the 2026-27 academic year, and the historical frequency of questions from the provided past year papers (PYQs), here is a highly probable predicted full CBSE Class 12 Chemistry Board Paper for 2027 [1, 2].

The paper consists of 33 questions divided into five sections (A, B, C, D, and E) for a total of 70 marks [3, 4].

PREDICTED CBSE CLASS 12 CHEMISTRY PAPER 2027

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

This section tests fundamental concepts, named reactions, and basic properties [5].

Q1. Which of the following pairs will form an ideal solution? (a) Chloroform and acetone (b) Ethanol and acetone (c) n-hexane and n-heptane (d) Phenol and aniline * **Chapter:** Solutions | **Confidence:** Very High [6]

Q2. The limiting molar conductivity (Λ_m°) for a weak electrolyte like CH_3COOH can be calculated using Kohlrausch's law if the values of the following are given: (a) $\Lambda_m^\circ \text{HCl}$, $\Lambda_m^\circ \text{KCl}$ and $\Lambda_m^\circ \text{CH}_3\text{COOK}$ (b) $\Lambda_m^\circ \text{NaCl}$, $\Lambda_m^\circ \text{KCl}$ and $\Lambda_m^\circ \text{CH}_3\text{COONa}$ (c) $\Lambda_m^\circ \text{HCl}$, $\Lambda_m^\circ \text{NaCl}$ and $\Lambda_m^\circ \text{CH}_3\text{COONa}$ * **Chapter:** Electrochemistry | **Confidence:** High [7, 8]

Q3. The slope in the plot of $[R]$ vs. time for a zero-order reaction is: (a) $\frac{+k}{2.303}$ (b) $-k$ (c) $\frac{-k}{2.303}$ (d) $+k$ * **Chapter:** Chemical Kinetics | **Confidence:** High [9]

Q4. Which of the following transition metal ions is colourless in an aqueous solution? (a) Ti^{4+} (b) Cr^{3+} (c) V^{3+} (d) Fe^{2+} * **Chapter:** d- and f-Block Elements | **Confidence:** Very High [10]

Q5. The counter ion in the coordination compound $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ is: (a) Ammine (b) Cobalt (c) Chloride (d) Nitro * **Chapter:** Coordination Compounds | **Confidence:** High [11]

Q6. Which of the following compounds is most reactive towards an $\text{S}_{\text{N}}1$ reaction? (a) $\text{CH}_2 = \text{CHCH}_2\text{Br}$ (b) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$ (c) $\text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br}$ (d) $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br}$ * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High [12]

Q7. Which of the following reactions is used to prepare salicylaldehyde from phenol? (a) Kolbe's reaction (b) Etard reaction (c) Reimer-Tiemann reaction (d) Stephen's reduction * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High [13, 14]

Q8. Iodoform test is **not** given by: (a) 2-pentanone (b) ethanol (c) ethanal (d) 3-pentanone * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High [15]

Q9. Arrange the following in the increasing order of their boiling points: $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $(\text{C}_2\text{H}_5)_3\text{N}$. * **Chapter:** Amines | **Confidence:** High [16, 17]

Q10. Which of the following is a polysaccharide? (a) Glucose (b) Maltose (c) Glycogen (d) Lactose * **Chapter:** Biomolecules | **Confidence:** Medium [18]

Q11. The reaction of alkyl chloride with AgF to yield alkyl fluoride is known as: (a) Finkelstein reaction (b) Swarts reaction (c) Wurtz reaction (d) Sandmeyer reaction * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High [19, 20]

Q12. What is the secondary valency of Cobalt in $[Co(en)_2Cl_2]^+$? (a) 6 (b) 4 (c) 2 (d) 8 * **Chapter:** Coordination Compounds | **Confidence:** High [21]

Directions for Q13 to Q16: Two statements are given—one labeled Assertion (A) and the other Reason (R). Select the correct answer from the codes (a), (b), (c), and (d).

Q13. Assertion (A): Conductivity decreases with a decrease in the concentration of the electrolyte. **Reason (R):** The number of ions per unit volume that carry the current in a solution decreases on dilution. * **Chapter:** Electrochemistry | **Confidence:** Very High [22]

Q14. Assertion (A): Tertiary amines are more basic than corresponding secondary and primary amines in the gaseous state. **Reason (R):** Tertiary amines have three alkyl groups which cause a +I (inductive) effect. * **Chapter:** Amines | **Confidence:** High [23]

Q15. Assertion (A): Chlorobenzene is less reactive towards nucleophilic substitution reactions compared to alkyl halides. **Reason (R):** The C–Cl bond in chlorobenzene acquires a partial double bond character due to resonance. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High [24]

Q16. Assertion (A): Elevation in boiling point is a colligative property. **Reason (R):** The lowering of the vapour pressure of a solution causes an elevation in the boiling point. * **Chapter:** Solutions | **Confidence:** High [25, 26]

SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

This section tests reasoning and brief numericals [5].

Q17. A first-order reaction takes 20 minutes for 50% completion. Calculate the time required for 90% completion of this reaction. ($\log 2 = 0.3010$) * **Chapter:** Chemical Kinetics | **Confidence:** Very High [27, 28]

Q18. Draw the structure and write the IUPAC name of the complex $[Co(en)_3]Cl_3$. What type of isomerism is shown by this complex? * **Chapter:** Coordination Compounds | **Confidence:** High [29, 30]

Q19. Give reasons for the following: (i) Aquatic species are more comfortable in cold water than in warm water. (ii) Why does water stop boiling when sugar is added to boiling water? * **Chapter:** Solutions | **Confidence:** High [31, 32]

Q20. Write the chemical equations involved in the following: (i) Hoffmann Bromamide Degradation reaction (ii) Gabriel Phthalimide Synthesis * **Chapter:** Amines | **Confidence:** Very High [33, 34]

Q21. Differentiate between S_N1 and S_N2 mechanisms on the basis of stereochemical outcomes. Why do primary alkyl halides prefer S_N2 ? * **Chapter:** Haloalkanes and Haloarenes

| **Confidence:** Very High [35, 36]

SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

This section tests mid-length derivations, multi-step conversions, and key numericals [37].

Q22. Calculate the maximum work (ΔG°) and $\log K_c$ for the given reaction at 298 K:
 $Ni_{(s)} + 2Ag_{(aq)}^+ \rightarrow Ni_{(aq)}^{2+} + 2Ag_{(s)}$ Given: $E_{Ni^{2+}/Ni}^\circ = -0.25\text{ V}$ and $E_{Ag^+/Ag}^\circ = +0.80\text{ V}$.
 ($1F = 96500\text{ C mol}^{-1}$) * **Chapter:** Electrochemistry | **Confidence:** Very High [38, 39]

Q23. Account for the following observations: (i) The atomic radii of **Zr** and **Hf** are almost identical. (ii) Transition metals and their compounds act as excellent catalysts. (iii) Mn^{2+} is much more resistant than Fe^{2+} towards oxidation. * **Chapter:** d- and f-Block Elements | **Confidence:** High [10, 40, 41]

Q24. Give simple chemical tests to distinguish between the following pairs of compounds: (i) Propanal and Propanone (ii) Phenol and Benzoic acid (iii) Ethanol and Phenol * **Chapter:** Organic Chemistry (Mixed) | **Confidence:** Very High [42-44]

Q25. (i) Write the mechanism of acid-catalysed hydration of an alkene to form an alcohol. (ii) Explain Williamson's ether synthesis. Why is it preferred for preparing mixed (unsymmetrical) ethers? * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** High [42, 45, 46]

Q26. State Henry's Law. Calculate the solubility of CO_2 in water at 298 K under 760 mm Hg. (K_H for CO_2 in water at 298 K is $1.25 \times 10^6\text{ mm Hg}$). * **Chapter:** Solutions | **Confidence:** Medium [32]

Q27. The rate of a reaction becomes four times when the temperature changes from 293 K to 313 K. Calculate the energy of activation (E_a) of the reaction assuming that it does not change with temperature. ($R = 8.314\text{ J K}^{-1}\text{ mol}^{-1}$, $\log 4 = 0.6021$) * **Chapter:** Chemical Kinetics | **Confidence:** Very High [47]

Q28. Differentiate between the following: (i) DNA and RNA (ii) Globular and Fibrous proteins (iii) Amylose and Amylopectin * **Chapter:** Biomolecules | **Confidence:** High [48, 49]

SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

These questions introduce a real-world or theoretical passage followed by sub-questions to test applied knowledge [50, 51].

Q29. Case Study 1: Solutions and Colligative Properties Boiling point or freezing point of a liquid solution is affected by the dissolved solids in the liquid phase. The addition of non-volatile substances to a solvent decreases the vapour pressure. The decrease in freezing point is directly correlated to the concentration of solutes dissolved in the solvent. This phenomenon is useful for applications such as freeze concentration of liquid food and to find the molar mass of an unknown solute [52]. (i) What is the Van't Hoff factor (i) for an ideal non-electrolyte solute? (1 Mark) (ii) Out of 1M glucose and 1M $MgCl_2$, which will have a

higher boiling point and why? (1 Mark) **(iii)** Calculate the mass of ascorbic acid (Molar mass = 176 g mol^{-1}) to be dissolved in 75 g of acetic acid to lower its freezing point by 1.5°C . ($K_f = 3.9 \text{ K kg mol}^{-1}$). (2 Marks) * **Chapter:** Solutions | **Confidence:** Very High [53]

Q30. Case Study 2: Carbonyl Compounds and Aldol Condensation The addition reaction of an enol or enolate to the carbonyl functional group of an aldehyde or ketone is known as aldol addition. The β -hydroxyaldehyde or β -hydroxyketone so obtained undergoes dehydration in the second step to produce a conjugated enone. Carbonyl compounds having an α -hydrogen undergo aldol condensation reactions [54]. **(i)** Provide the structure of the product formed when two molecules of ethanal react in the presence of dilute NaOH . (1 Mark) **(ii)** Why is the presence of an α -hydrogen essential for an aldol condensation? (1 Mark) **(iii)** Can benzaldehyde undergo an aldol condensation? If not, what alternate reaction does it undergo when treated with concentrated alkali? Write the equation. (2 Marks) * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High [54-56]

SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

These are comprehensive questions often split into sub-parts, testing deep conceptual clarity and extensive numerical solving [57].

Q31. (A) Electrochemistry Numericals & Concepts (i) State Kohlrausch's law of independent migration of ions. How does it help in calculating the limiting molar conductivity of a weak electrolyte? (2 Marks) (ii) The standard electrode potential for $\text{Sn}^{4+}/\text{Sn}^{2+}$ is $+0.15 \text{ V}$ and for Cr^{3+}/Cr is -0.74 V . Calculate the standard cell potential (E_{cell}°) when these two couples are connected. (1 Mark) (iii) Write the Nernst equation and calculate the emf of the following cell at 298 K: $\text{Mg}_{(s)}|\text{Mg}^{2+}(0.1 \text{ M})||\text{Cu}^{2+}(0.01 \text{ M})|\text{Cu}_{(s)}$ Given $E_{\text{cell}}^\circ = +2.71 \text{ V}$. (2 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High [58-60]

Q32. (A) d- and f-Block Elements (i) Describe the preparation of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) from chromite ore (FeCr_2O_4). Write the balanced chemical equations for the reactions involved. (3 Marks) (ii) What is the effect of increasing pH on a solution of potassium dichromate? Support your answer with a chemical equation. (2 Marks) * **OR (B)** (i) Write the complete ionic equations for the oxidation of Fe^{2+} to Fe^{3+} by dichromate ions ($\text{Cr}_2\text{O}_7^{2-}$) in an acidic medium. (2 Marks) (ii) Account for the following: a) Transition metals form complex compounds easily. b) Ce^{4+} is a strong oxidizing agent. c) The highest oxidation state is exhibited in oxoanions of a transition metal. (3 Marks) * **Chapter:** d- and f-Block Elements | **Confidence:** High [61-64]

Q33. (A) Aldehydes, Ketones and Carboxylic Acids An organic compound (A) with the molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an orange-red precipitate with 2,4-DNP reagent and gives a yellow precipitate on heating with iodine in the presence of sodium hydroxide. It neither reduces Tollens' reagent nor Fehling's solution. On drastic oxidation with chromic acid, it gives a carboxylic acid (B) having the molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Identify the compounds (A) and (B) and write the equations for the reactions involved. (5 Marks) * **OR (B)** Complete

the following reactions: (i) $\text{CH}_3 - \text{CO} - \text{CH}_3 \xrightarrow{\text{Zn(Hg)/conc.HCl}} ?$ (Clemmensen Reduction)
(ii) $\text{C}_6\text{H}_5\text{CHO} + \text{H}_2\text{N} - \text{NH}_2 \xrightarrow{\text{H}^+} ?$ (iii) $\text{CH}_3\text{COOH} \xrightarrow{\text{Cl}_2/\text{Red P}_4} ?$ (HVZ Reaction) (iv)
 $2\text{HCHO} \xrightarrow{\text{Conc.KOH}} ?$ (Cannizzaro Reaction) (v) $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow{\text{PCC}} ?$ * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High [65-68]

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.